OPEN-LOOP AND CLOSED-LOOP CONTROL UNDER LOAD

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Identify the controller, actuator, process, sensor, reference, disturbance and feedback path on the ACE-Lab DC motor rig.	☐
2	Run the motor in open loop and in closed loop, and apply the same load disturbance to each.	☐
3	Compare the measured responses and explain, in terms of feedback, why the closed-loop system holds the motor speed when a load is applied.	☐

2.0 Hardware Set-Up

This practical exercise builds on the hardware and algorithm set-up described in the practical exercise, DC Motor Speed Measurement using an Encoder, which details the operation of a DC motor with an encoder and the measurement of motor speed in revolutions per minute (RPM). The set-up is not repeated here.

The closed-loop part of this exercise uses the PI speed controller from the practical exercise, DC Motor Speed Control Using a PI Controller. A load is applied by pressing lightly on a disc fixed to the motor shaft. A disc of card or plastic, about 40 mm in diameter, can be pushed onto the shaft or attached with a small hub.

Safety: apply the load with a fingertip or the end of a pencil eraser on the face of the disc, never by gripping the shaft. Keep loose cables and hair clear of the shaft while the motor is running.

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. The motor is first run in open loop, where a fixed PWM value is applied regardless of the speed, and then in closed loop, where a controller adjusts the PWM value using the measured speed. The same load is applied in both cases so that the two responses can be compared directly. The exercise supports the ACE-CORE learning guide, Introduction to Control Systems.

Background: Open-Loop and Closed-Loop Control

In open-loop control, shown in Figure 1, the input to the process is set in advance and does not depend on the output. Any disturbance that acts on the process, such as a change in load, changes the output and nothing corrects it.

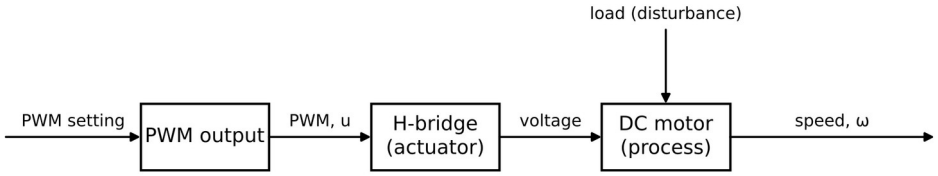


Figure 1: *Open-Loop Speed Control of the DC Motor*

In closed-loop control, shown in Figure 2, the output is measured and compared with the desired value, the setpoint. The controller acts on the difference between them, the error, so a disturbance that changes the output also changes the error, and the controller responds to correct it.

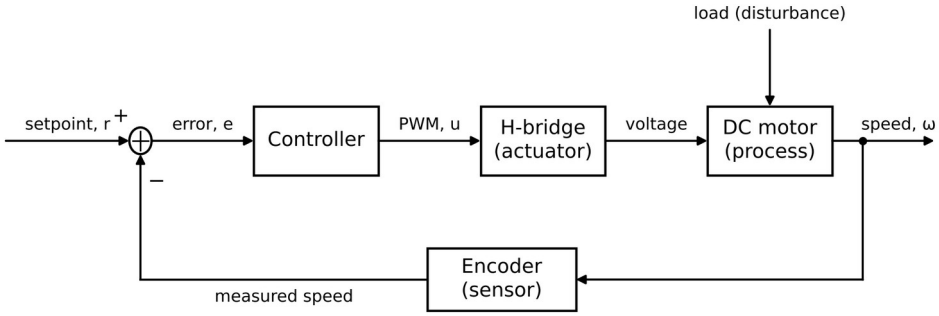


Figure 2: *Closed-Loop Speed Control of the DC Motor*

Identify the Parts of the Control System

Before running any tests, identify each part of the closed-loop system on your own ACE-Lab hardware and in the Simulink model, and complete Table 2. Describe each part in terms of what it is physically on the rig or in the model, not only its name in the block diagram.

Table 2: *Parts of the Closed-Loop Speed Control System on the ACE-Lab Rig*

Part	On your ACE-Lab rig or Simulink model
Reference (setpoint)	
Controller	
Actuator	
Process	

Part	On your ACE-Lab rig or Simulink model
Sensor	
Disturbance	
Feedback path	

Run the Motor in Open Loop under Load

In the open-loop model, a Constant sets the voltage applied to the motor and a Gain of 255/5 converts it into the PWM value, see Figure 3. Nothing in the model uses the measured speed to change the input.

[Screenshot to insert] *Open-loop model: Constant (3 V) → Gain 255/5 → PWM output; encoder speed (RPM) → Scope. Scopes log u, y and t.*

Figure 3: *Simulink Model for Open-Loop Speed Control*

1. Open the speed measurement model from the practical exercise, DC Motor Speed Measurement using an Encoder, and save it under a new name.
2. Set the Constant driving the PWM output to 3 V, through a Gain of 255/5. Log the applied voltage, the measured speed and the time to the workspace as *u*, *y* and *t*, using the Save format Structure With Time.
3. Set the Stop Time to 12 s and run the model on the hardware. Allow the speed to settle for about 3 s.
4. Apply a light, steady load to the disc for about 3 s, then release it. Repeat the load once more before the run ends, trying to press with the same force each time.
5. Record the steady speed before the load and the lowest speed during the load in Table 3.

Run the Motor in Closed Loop under Load

Use the closed-loop model from the practical exercise, DC Motor Speed Control Using a PI Controller, with the setpoint equal to the steady open-loop speed you recorded. The two tests then start from the same speed, and any difference in the response to the load is due to the feedback.

1. Open the PI speed control model and set the speed setpoint to the steady open-loop speed from Table 3.
2. Log the setpoint, the measured speed, the PWM command (in volts, after a Gain of 5/255) and the time.
3. Run the model for 12 s and apply the load in the same way as before.

4. Record the speed before the load, the lowest speed during the load, the speed after recovery, and the applied voltage before and during the load in Table 3.

Example results are shown in Figure 4. They are simulated with the motor model identified in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares (K = 28 RPM/V, τ = 0.15 s), with the load represented as a loss of 1.2 V at the motor, so they show the behaviour to expect rather than measured data. In open loop the speed falls by about 34 RPM and stays low until the load is removed. In closed loop the speed dips briefly and returns to the setpoint, because the controller raises the applied voltage to make up for the load.

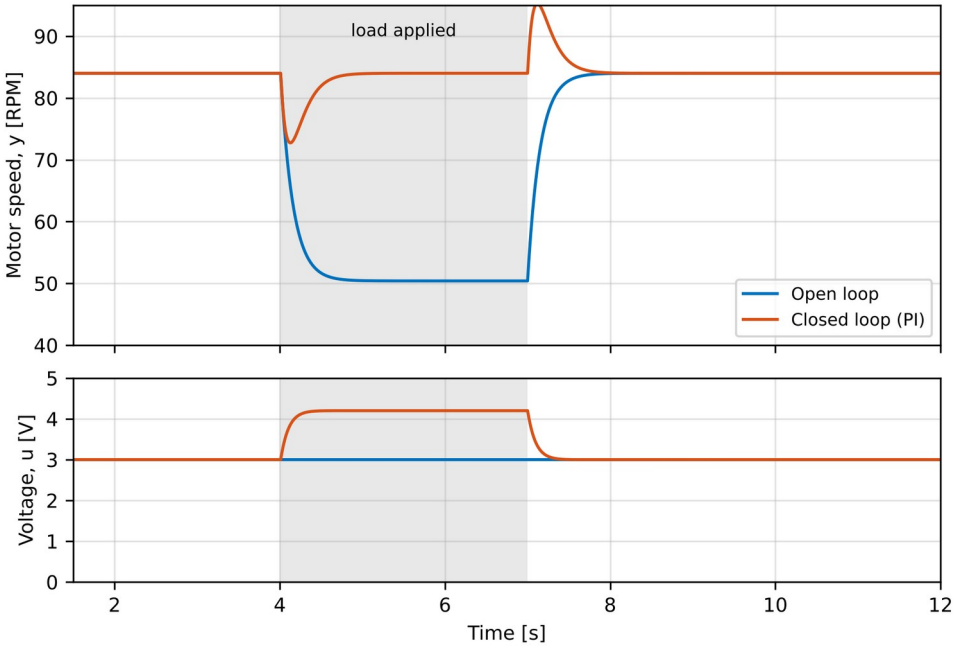


Figure 4: *Simulated Example of the Motor Speed (Top) and Applied Voltage (Bottom) in Open Loop and Closed Loop with a Load Applied from 4 s to 7 s*

Table 3: *Open-Loop and Closed-Loop Results for Your Motor*

Test	Speed before load (RPM)	Lowest speed (RPM)	Speed during load after 2 s (RPM)	Voltage before / during load (V)
Open loop				3.0 / 3.0
Closed loop				

Plot the open-loop and closed-loop speeds on the same axes, and plot the closed-loop voltage beneath them, as in Figure 4. Use your results to answer the following questions.

1. Which signal did the controller change to reject the load, and by how much did it change?
2. What would happen in closed loop if the encoder cable were disconnected while the motor was running? Explain why, using Figure 2.

3. Give one application where open-loop speed control would be good enough, and one where closed-loop control is essential.

4.0 Concluding Remarks

This practical exercise compared open-loop and closed-loop speed control of a DC motor under the same load disturbance. Using MATLAB, Simulink and ACE-Lab hardware, the parts of a feedback control system were identified on a real rig, and the response of each configuration to a load was measured.

In open loop, the load reduced the speed for as long as it was applied, because nothing in the system measured the speed. In closed loop, the encoder measured the drop in speed, the controller increased the applied voltage, and the speed returned to the setpoint. This ability to reject disturbances, and to compensate for changes in the process itself, is the main reason feedback control is used.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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